Combined Probabilities

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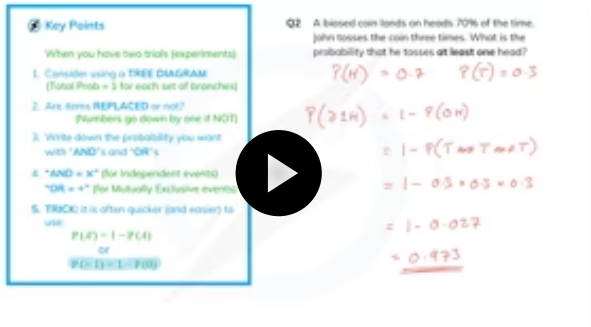

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Combined probability

How do I calculate combined probabilities?

- You can calculate probabilities of **one event after another** without needing tree diagrams
  - These are called **combined** (or successive) probabilities
- There are two **rules** to learn
  - **And** means **multiply** and **or** means **add**
  - P(A **and** B) = P(A) x P(B)
  - P(AA **or** BB) = P(AA) + P(BB)
- Try to **rephrase** each question using and / or
  - For example, when flipping a coin twice:
    - P(two heads) = P(head **and** head)
    - P(both the same) = P(head **and** head **or** tail **and** tail) = P(HH) + P(TT)
- Remember that P(**not** A) = 1 - P(A)

What does independent mean?

- **Independent events** are events that **do not affect** each other
  - e.g. the probability of rolling a 6 on a fair dice and the probability of getting a head when flipping a coin

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

not replacing it, then finding the probability of taking a second red card out of the same pack

- The first event **affected** the number of cards left for the second event

Worked Example

A box contains 3 blue counters and 8 red counters.
A counter is taken at random and its colour is noted.
The counter is put back into the box.
A second counter is then taken at random, and its colour is noted.

Work out the probability that

(a) both counters are red,

Answer:

P(both red) = P(red and red)
This is P(red) × P(red) using the 'and rule'

$$\frac{8}{11} \times \frac{8}{11}$$

$$\frac{64}{121}$$

(b) the two counters are of different colours.

Answer:

P(different colours) = P(blue and red or red and blue)
This is P(B and R) + P(R and B) using the 'or rule'
This is P(B) × P(R) + P(R) × P(B) using the 'and rule' twice

$$\frac{8}{11} \times \frac{3}{11} + \frac{3}{11} \times \frac{8}{11}$$

$$= \frac{24}{121} + \frac{24}{121}$$

$$\frac{48}{121}$$

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